Crop Suitability Classification Using Foundational Machine Learning Models

Track 1 Final Capstone — Problem 19 | Learn Depth Academy LLP
Author: Charan
September 2026

Abstract

Choosing a crop that suits local soil and climate is a key decision for farmers, and a wrong choice lowers yield and wastes inputs. This project builds an end-to-end, zero-cost machine learning solution that recommends the most suitable of 22 crops from seven measurable inputs: soil nitrogen (N), phosphorus (P), potassium (K), soil pH, temperature, humidity and rainfall. Using the public Crop Recommendation Dataset (2,200 samples), we performed data quality checks and exploratory analysis, built a leakage-safe preprocessing pipeline, and compared three foundational classifiers — Logistic Regression, K-Nearest Neighbours (KNN) and a Decision Tree — tuned with 5-fold stratified cross-validation. All three models exceeded 97% accuracy on a held-out test set. The Decision Tree was selected for its best cross-validated macro-F1 (0.986) and full interpretability, and achieved 97.95% test accuracy and 0.979 macro-F1, with only 9 errors out of 440 test samples, all between agronomically similar crops. Humidity, phosphorus and rainfall were the most influential features. The model is deployed in a lightweight Streamlit application that validates inputs and returns a crop recommendation with a confidence score.

1. Introduction

Agriculture supports a large share of India's population, yet crop selection is often based on tradition rather than measured soil and weather conditions. Soil testing (through programmes such as Soil Health Cards) and weather data are increasingly available, which creates an opportunity to support crop choice with data-driven tools.

Machine learning can learn the relationship between soil/climate conditions and the crops that grow well under them. This project investigates whether simple, foundational classifiers — which are fast, cheap to run and easy to explain — are sufficient for this task, and demonstrates the result through a working prototype.

2. Problem Definition

Task: Multi-class classification. Given a new observation of seven soil and climate measurements, predict the most suitable crop category from 22 classes.

Inputs: N, P, K (soil nutrient ratios), temperature (°C), humidity (%), pH, rainfall (mm).

Output: The predicted crop label, with a probability (confidence) score and top alternatives.

Research question: Can a foundational classifier reliably distinguish crop-suitability categories, and which soil/climate characteristics matter most?

Constraints: Free/open-source tools only (Python, scikit-learn, pandas, Streamlit); educational prototype, not a production system.

3. Related Work

Liakos et al. (2018) reviewed machine learning applications in agriculture and found crop management — including yield prediction and crop recommendation — to be one of the most active areas, with tree-based models, support vector machines and neural networks widely used. van Klompenburg, Kassahun and Catal (2020) systematically reviewed crop yield prediction studies and reported that temperature, rainfall and soil type are the most commonly used features, and that relatively simple models often perform competitively.

The Crop Recommendation Dataset used here (Ingle, 2020) is a widely used public benchmark for crop recommendation; many public notebooks report high accuracy on it with tree-based models. This project follows that line of work but deliberately restricts itself to foundational models, focuses on leakage-safe evaluation, and adds error analysis and a deployable prototype.

4. Dataset

The Crop Recommendation Dataset (Ingle, 2020, Kaggle) contains 2,200 rows and 8 columns: seven numeric features and one target label. It was compiled by augmenting Indian rainfall, climate and fertiliser data.

Table 1. Dataset features and summary statistics

Feature	Description	Min	Mean	Max
N	Nitrogen ratio in soil	0	50.6	140
P	Phosphorus ratio in soil	5	53.4	145
K	Potassium ratio in soil	5	48.1	205
temperature	Temperature (°C)	8.8	25.6	43.7
humidity	Relative humidity (%)	14.3	71.5	100.0
ph	Soil pH	3.5	6.5	9.9
rainfall	Rainfall (mm)	20.2	103.5	298.6
label	Crop (22 classes)	—	—	—

The 22 crops are: apple, banana, blackgram, chickpea, coconut, coffee, cotton, grapes, jute, kidney beans, lentil, maize, mango, moth beans, mung bean, muskmelon, orange, papaya, pigeon peas, pomegranate, rice and watermelon.

5. Methodology

The project followed the five-stage workflow prescribed in the brief: Research → Data + EDA → Model → Evaluate → Prototype + Paper.

- **Data quality:** checked data types, missing values, duplicate rows, duplicate feature vectors and class balance.
- **EDA:** distributions and skewness, IQR-based outlier detection, correlation analysis and per-crop feature profiles.

- **Split:** stratified 80/20 train/test split (1,760 / 440 rows), made before any model fitting, with a fixed random seed (42) for reproducibility.
- **Leakage control:** feature scaling (StandardScaler) is placed inside a scikit-learn Pipeline, so it is fitted only on training folds. A check confirmed zero identical rows between train and test sets.
- **Modelling:** Logistic Regression, KNN and Decision Tree, each tuned by grid search with 5-fold stratified cross-validation on the training set, optimising macro-F1.
- **Model selection:** based on the cross-validated score, not the test score, so that the test set remains an unbiased estimate.
- **Evaluation:** accuracy, macro precision, recall, F1-score, confusion matrix and one-vs-rest ROC-AUC on the held-out test set, plus error analysis and permutation feature importance.
- **Deployment:** the full fitted Pipeline is saved with joblib and loaded by the Streamlit app, guaranteeing the same preprocessing at prediction time.

6. Exploratory Data Analysis

6.1 Data quality

The dataset has no missing values, no duplicate rows and no duplicate feature vectors. Every crop has exactly 100 samples, so the classes are perfectly balanced (Figure 1). As a result, accuracy is a fair headline metric and no resampling is needed.

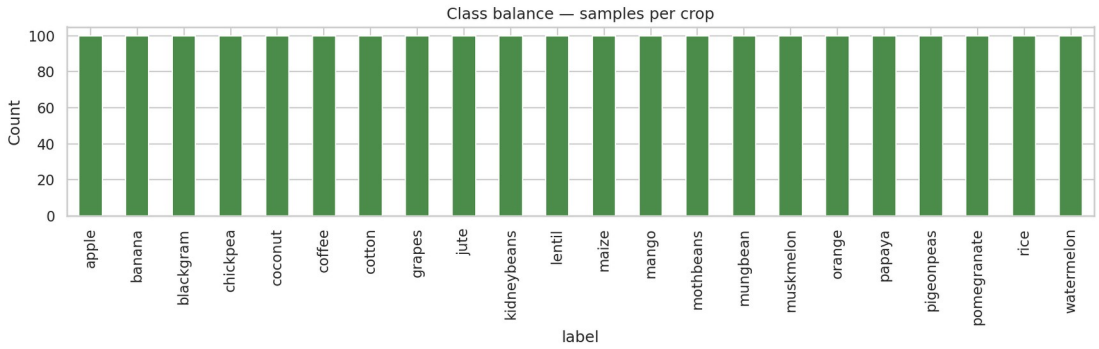


Figure 1. Samples per crop — perfectly balanced classes

6.2 Distributions and outliers

K (skew 2.38), P (1.01) and rainfall (0.97) are right-skewed, while humidity is left-skewed (−1.09). The IQR rule flags outliers in P (138), K (200) and rainfall (100). On inspection, these values come entirely from specific crops: all high-K values and most high-P values belong to grapes and apple, and high rainfall values belong to rice, papaya and coconut. These are genuine agronomic signals, not errors, so they were kept.

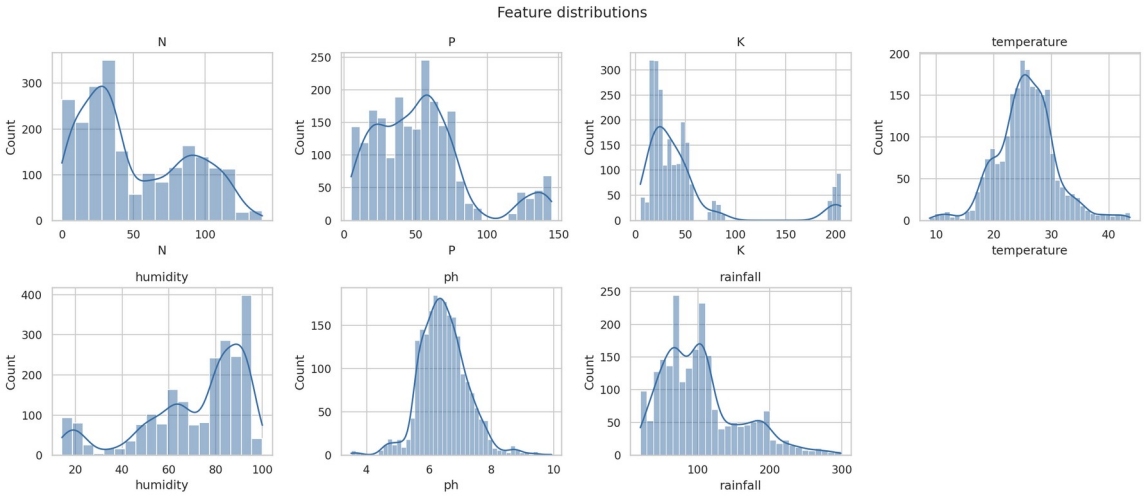


Figure 2. Distribution of each feature

6.3 Relationships

Most features are weakly correlated with each other. The only strong correlation is between P and K (r ≈ 0.74), driven by the grapes and apple classes. Per-crop profiles (Figure 3) show that each crop has a distinctive combination of conditions — for example, rice needs high rainfall, chickpea and kidney beans grow in low humidity, and grapes and apple need very high P and K. This suggested early on that the classes would be highly separable.

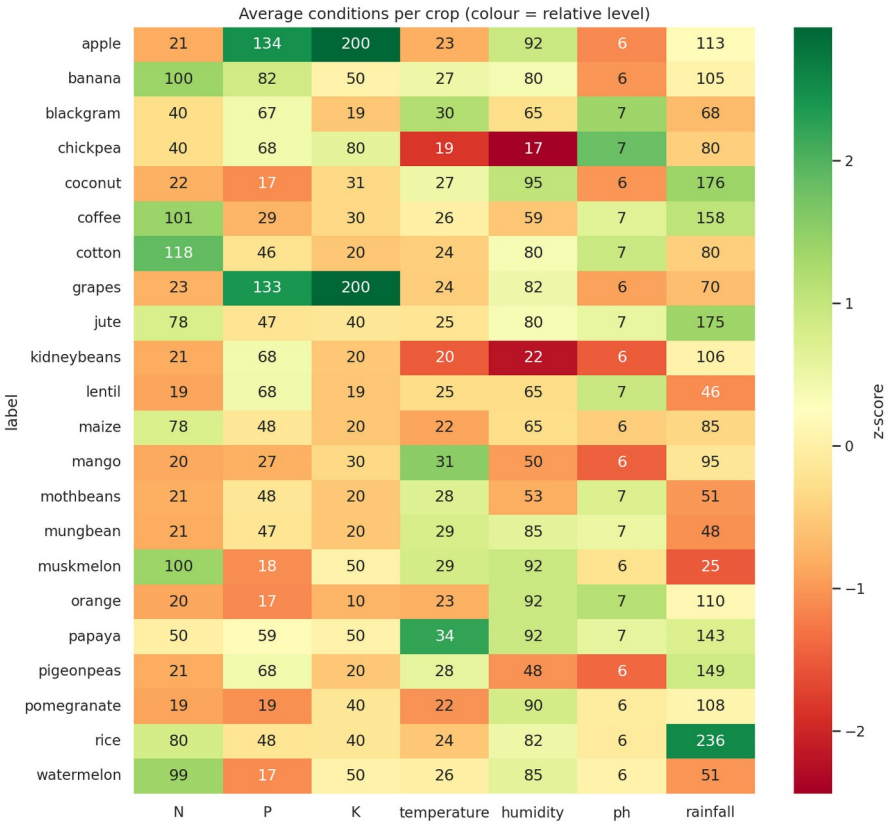


Figure 3. Average conditions per crop (numbers = mean values; colour = relative level)

7. Model Development

Three foundational classifiers were chosen because they represent three different learning approaches and are easy to explain:

- **Logistic Regression** (linear, multinomial) — a baseline that assumes classes can be separated by linear boundaries. Requires scaled inputs. Tuned: regularisation strength C ∈ {0.1, 1, 10, 100}.
- **K-Nearest Neighbours** — predicts from the most similar training samples. Distance-based, so scaling is essential. Tuned: k ∈ {3, 5, 7, 9, 11}, uniform or distance weighting.
- **Decision Tree** — learns human-readable if/then rules. Does not need scaling. Tuned: max depth ∈ {5, 10, 15, none}, minimum samples per leaf ∈ {1, 2, 5}.

The best settings found were C = 100 for Logistic Regression, k = 5 with distance weighting for KNN, and unlimited depth with at least 2 samples per leaf for the Decision Tree (final tree: depth 17, 37 leaves). The tree's first splits are easy to read: rainfall ≤ 30 mm identifies muskmelon; among the rest, humidity ≤ 28% separates kidney beans and chickpea (split on K); and P > 107.5 isolates grapes and apple.

8. Results

Table 2. Model comparison (CV = 5-fold cross-validation on training set; other metrics on the 440-row test set, macro-averaged)

Model	CV F1	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.978	0.984	0.985	0.984	0.984	1.000
KNN (k = 5)	0.968	0.982	0.983	0.982	0.982	0.999
Decision Tree (selected)	0.986	0.980	0.981	0.980	0.979	0.989

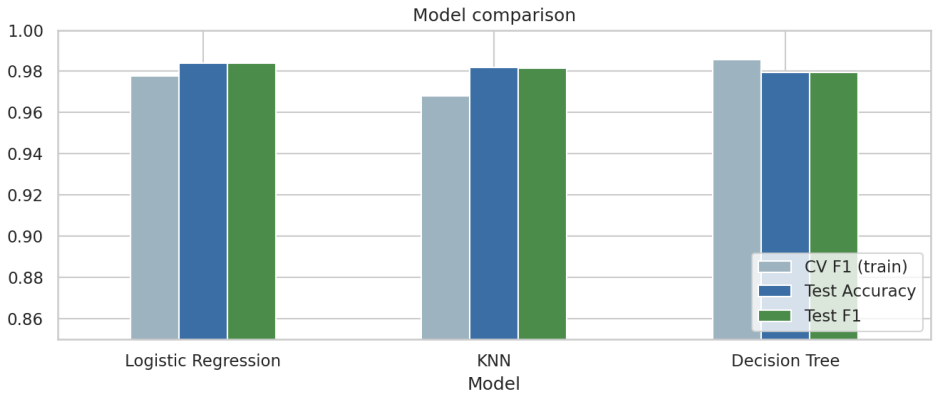


Figure 4. Cross-validated and test performance of the three models

Final model choice. The Decision Tree had the highest cross-validated macro-F1 (0.986), which is the criterion we fixed before looking at test results. Its test performance (97.95% accuracy) is within 0.5 percentage points of the other models — a difference of about two test samples, which is within normal variation. It was chosen because it (a) scored best on the pre-agreed selection criterion, (b) is fully interpretable, as its rules can be printed and explained to a farmer or examiner, and (c) needs no scaling and predicts instantly. Logistic Regression is a close alternative and is noted for future comparison.

8.1 Confusion matrix and error analysis

The Decision Tree classified 20 of the 22 crops with 100% precision and recall. It made only 9 errors in 440 test samples (Figure 5). The errors are concentrated in two groups of crops that have very similar growing conditions:

- **Pulses — blackgram, lentil, moth beans (7 errors):** all three have low K (~19–20), moderate P and 45–68 mm rainfall. Blackgram had the lowest recall (0.80).
- **Rice ↔ jute (2 errors):** nearly identical N, P, K, temperature and humidity; they differ mainly in rainfall (mean 236 mm vs 175 mm), and the misclassified samples fall in the overlap zone.
- **Blackgram → maize (1 error):** a blackgram sample with unusually high N (60) for its class.

These errors make agronomic sense: they occur between crops that could plausibly be grown under the same conditions, so they are less harmful than they look.

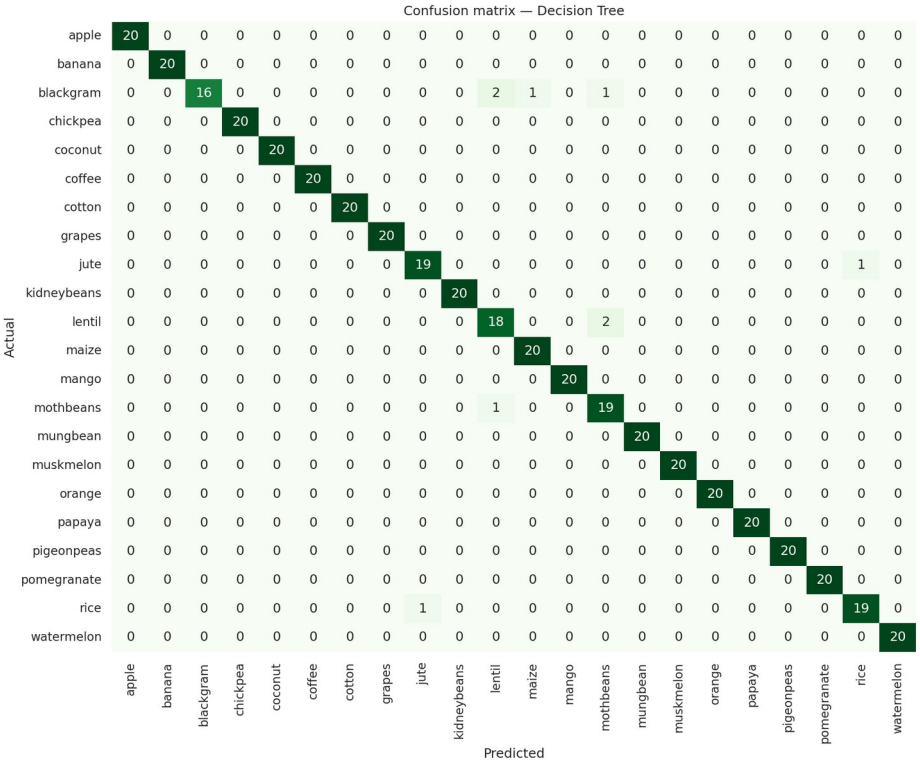


Figure 5. Confusion matrix of the Decision Tree on the test set

8.2 Feature relevance

Permutation importance (the drop in test accuracy when a feature is randomly shuffled) shows that humidity (0.58), phosphorus (0.47) and rainfall (0.42) are the most important features, followed by nitrogen (0.28) and potassium (0.11). Temperature (0.03) and pH (0.01) contribute little to this model, because the other features already separate the crops.

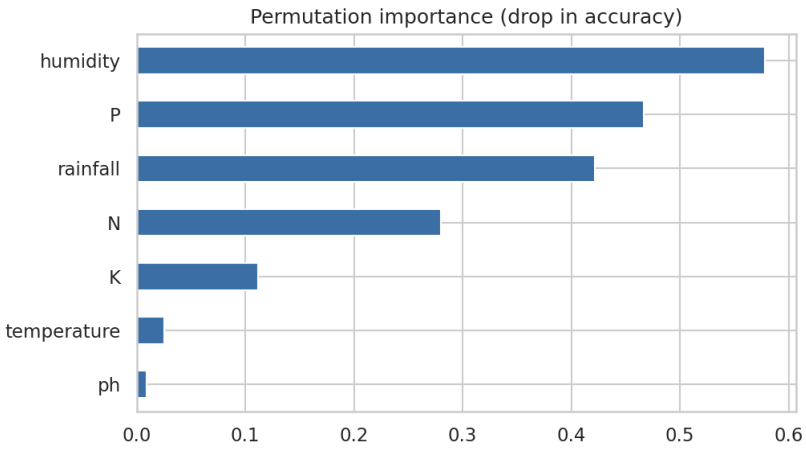


Figure 6. Permutation feature importance

9. Prototype Application

A Streamlit web application (app.py) demonstrates the model. The user enters the seven values; the app:

- validates inputs against physically sensible limits (for example, humidity 0–100%, pH 0–14);
- warns when a value falls outside the range seen in training, so the prediction may be less reliable;
- passes the input through the saved Pipeline, applying exactly the same preprocessing as in training;
- displays the recommended crop, a confidence score and the top alternatives.

The app was tested with known samples; for example, a typical rice input (N 90, P 42, K 43, 20.9 °C, 82% humidity, pH 6.5, 203 mm rainfall) correctly returns Rice, and a high-P/high-K input returns Apple.

10. Discussion

The results show that foundational models are sufficient for this problem: all three models exceed 97% accuracy. This is because the dataset's classes have well-separated soil and climate profiles, as the EDA showed. The Decision Tree's rules closely match agronomic knowledge — rainfall, humidity and nutrient levels define each crop's niche.

However, very high accuracy on a clean, balanced benchmark should be interpreted carefully. The dataset appears to be generated or augmented from typical ranges per crop, with exactly 100 samples each, so real-world field data is likely to be noisier and more overlapping. The remaining errors between the pulses show where the model would need extra information, such as season, soil type or region.

11. Limitations

- Single public dataset with uniform class sizes; it may not reflect real farm-level variability.
- No location, season, soil texture, irrigation or market-price information.
- Covers only 22 crops; the model cannot recommend a crop outside this list.
- The prediction says which crop best matches the conditions, not the expected yield or profitability.

- Educational prototype — not validated with agronomists or field trials, and not intended as agronomic advice.

12. Future Scope

- Validate on real Indian soil health card and weather data (for example, data.gov.in and IMD).
- Add season (Kharif/Rabi/Zaid), soil type and district as features.
- Compare with ensemble methods (Random Forest, Gradient Boosting) and calibrate probabilities.
- Return a ranked list with fertiliser adjustment suggestions (for example, how much N to add for a target crop).
- Deploy on Streamlit Community Cloud and add regional-language support for farmers.

13. Conclusion

This project delivered an end-to-end, zero-cost crop suitability classifier. After confirming the data was clean and balanced, and exploring each crop's distinctive soil and climate profile, three foundational models were compared under a leakage-safe, cross-validated setup. The selected Decision Tree reached 97.95% test accuracy and 0.979 macro-F1, with its few errors limited to agronomically similar crops, and it provides transparent rules that can be explained. Humidity, phosphorus and rainfall were the strongest predictors. The Streamlit prototype turns the model into a simple tool that validates inputs and returns a clear crop recommendation. The work answers the research question positively: a foundational classifier can reliably distinguish crop-suitability categories on this dataset.

References

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